Unit 3: Research Design and Instrumentation

Research Design, Type of research and research design, Descriptive Research, Relational Research and Experimental Research, Relationship Between Descriptive Research, Relational Research, and Experimental Research, Measuring the human, Preparing research design: universe, population and sample size, Sample selection and instrument design, Basics of experimental research, Experimental design, Online research design, Use of qualitative methods in CA: diaries, case studies, interviews and focus groups, Ethnography: interviewing, observing, analyzing, repeating, and theorizing, Automated data collection methods.

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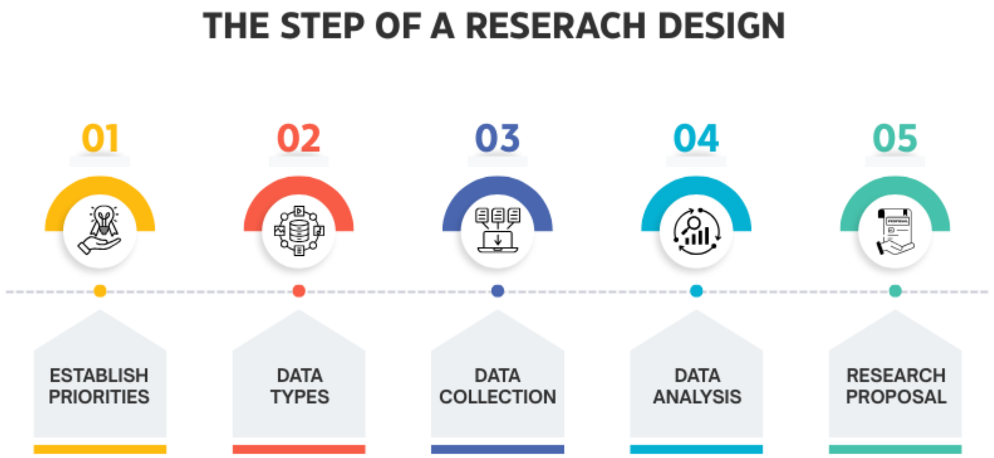
Research Design

- A research design is a type of framework developed in the planning stages to help and guide the research in a suitable direction, including determining the research questions, selecting appropriate research methods, and identifying suitable data collection and analysis methods.
- A well-structured plan helps generate valid, reliable, and accurate results.



Research Design:Elements

A good research design typically has the following elements:

- **Clear objective**: The research question must be stated in clear terms.
- **Sampling**: The size, methods, and inclusion and exclusion criteria should be clearly specified.
- **Data collection and analysis**: These methods should be suitable for the type of research methodology chosen for your study.
- **Research methodology**: The overall approach of your research should be described for readers.
- **Study duration**: To better understand the research context, the study duration should be specified, along with the time required for data collection and analysis.
- **Ethical considerations**: Important ethical information, such as informed consent and confidentiality, should be included to ensure research integrity.

Research Design

Purpose of Research Design

- The primary purpose of a research design is to create a study plan that helps assess the cause-and-effect relationships between variables.
- To create a roadmap for the study, which researchers can refer to for tracking their progress.
- Ensure alignment between research objectives and results.
- Minimize errors in data collection and analysis by providing guidelines.
- Help other researchers to replicate the findings.
- Ensure the validity and accuracy of the results and lend credibility to the study.

Research Design

Key Components of Research Design

- A research design has several important components, and it is essential to include these to develop a well-structured and complete design.
- **Research question and objectives**: Help guide the research process
- **Variables**: The concepts whose relationships need to be studied
- **Sampling techniques**: Suitable samples of appropriate sizes should be used depending on the study type.
- **Data collection methods**: Should be selected based on the study, and can include surveys, interviews, observations, or experiments.
- **Data analysis methods**: These methods should be based on the type of data collected.

Research Design

Characteristics of Research Design

For ensuring academic integrity and authenticity, all research designs should have the following characteristics:

- **Neutrality**: The research should present an unbiased perspective, free of any preconceived notions. Researchers' expectations should not affect the reporting of the findings.
- **Reliability**: The research should produce consistent results over repeated measures and there should be few errors due to chance.
- **Validity**: Refers to the minimization of nonrandom errors.
- **Generalizability**: The research results should be applicable to a larger population and not just the study sample.

Research Design

How to Choose the Right Research Design

- **Research question**: The type of research question and study objectives determines the type of research design most appropriate for your research. For instance, if you're studying consumers' purchasing behaviors, a qualitative design would be suitable. However, if you're testing hypotheses, you may need numerical data and therefore a quantitative design would be more appropriate.
- **Data availability**: The presence of a large, representative sample of data can be helpful in a quantitative design, and a smaller sample may be more appropriate for a qualitative design.
- **Resources and time availability**: The amount of resources and time available will determine the type of research design required. Experimental designs, for instance, can be time- and resource-intensive, so if you're short on either time or resources, then this design type won't be suitable.

Research Design

Benefits of Research Design

An effective research design can be beneficial to researchers in several ways. Listed below are a few key benefits of research design.

- Ensures that the research question and objectives are clearly defined.
- Helps researchers identify potential limitations.
- Helps identify suitable data collection methods and eventually saves time and resources.
- Makes it easier for other researchers to repeat the research and validate the findings.
- Improves the credibility of the research.

Framework of Research Design

From the preceding points, one thing is clear that the design actually facilitates the research toward a right direction.

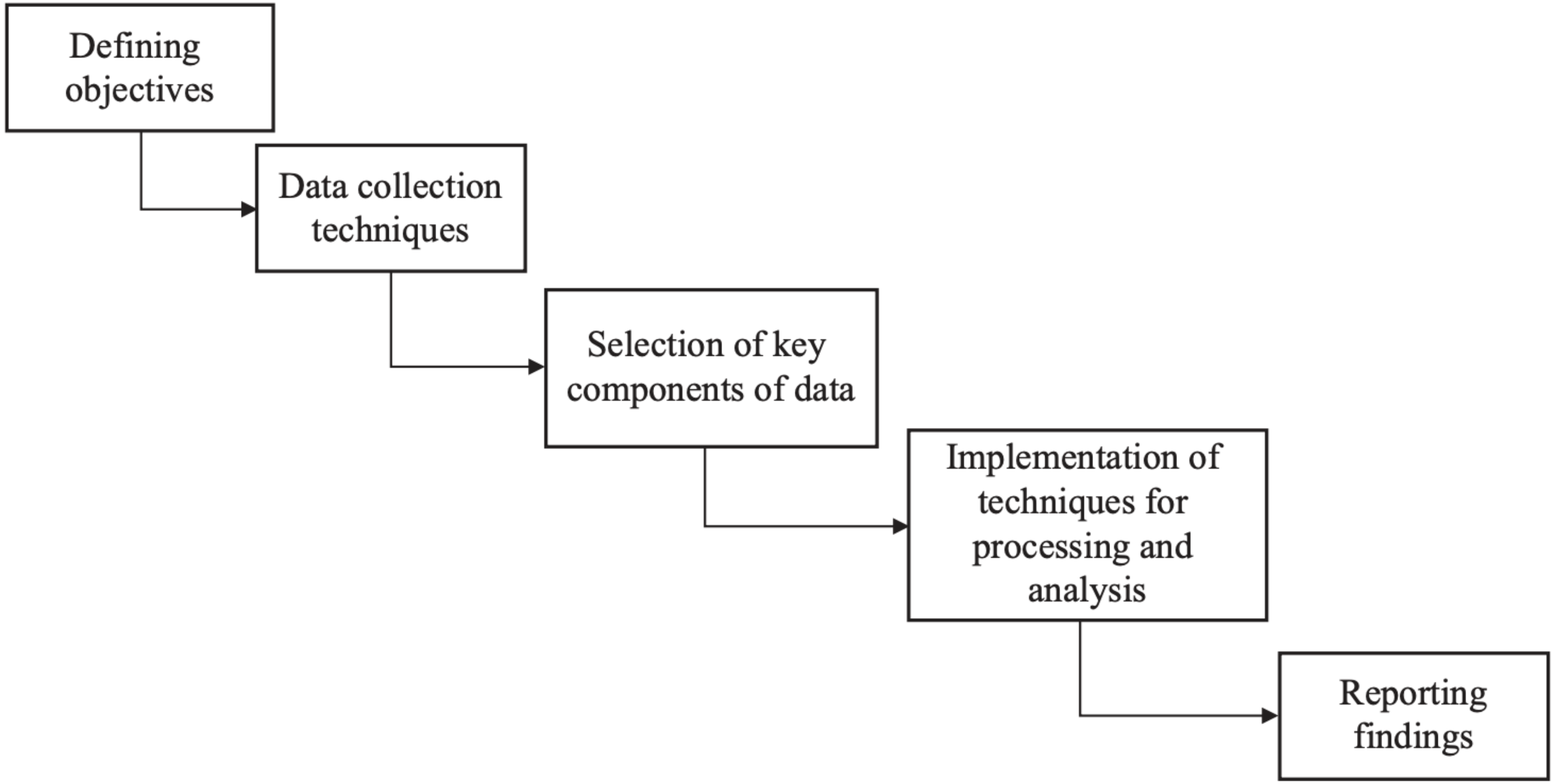
So essentially a research framework deals with:

- Defining precise, up to the point problem statement
- Approaches/techniques that can be put to practice to collect samples/data
- The details about the data that need to be analyzed and further researched upon
- Approaches/experimental setups to be executed for the data processing and analysis.

Framework of Research Design

- One thing that a researcher need to keep in mind is that all types of research designs are not applicable to all types of research problems and vice-versa is also not true.
- The type of research study would be the one determining the design for it.
- In general, without belonging to any specific category of research, the framework is constituted of components as

Framework of Research Design



Key Concept and terminologies

Variables

- A **variable** is any characteristic, quantity, or attribute in research that can take different values.
- Variables are the basic elements of any research study. They help researchers measure, analyze, and understand relationships.

Types of Variables

- **Quantitative Variables:** Variables that can be measured numerically.
  Example: Temperature (°C), earthquake magnitude, height, weight
- **Qualitative (Categorical) Variables :** Variables that describe qualities or categories rather than numbers.
  Example: Air quality (Good/Bad), personality (Introvert/Extrovert)

Key Concept and terminologies

- **Continuous Variables:** Variables that can take any value within a range: *Height, weight, temperature*
- **Categorical Variables:** Variables grouped into categories or classes. *Blood group (A, B, AB, O)*

Key Concept and terminologies

Independent and Dependent Variables

- **Independent Variable (IV) :** The variable that is manipulated or controlled by the researcher to observe its effect.
- **Dependent Variable (DV) :** The variable that is measured and changes as a result of the independent variable.

Example

- Study: Effect of study hours on exam marks
  - Independent Variable: Study hours
  - Dependent Variable: Exam marks
- Study: Relationship between height and weight
  - Height → Independent
  - Weight → Dependent

Key Concept and terminologies

Independent and Dependent Variables

Fertilizer and Plant Growth

Study Topic: Effect of fertilizer on plant growth

- **Independent Variable (IV):** Amount/type of fertilizer
- **Dependent Variable (DV):** Plant growth (height, number of leaves)

Social Media Use and Academic Performance

Study Topic: Effect of social media usage on students

- **Independent Variable (IV):** Time spent on social media
- **Dependent Variable (DV):** Academic performance (grades)

Key Concept and terminologies

Independent and Dependent Variables

Website Loading Speed and User Satisfaction

Study Topic: Effect of website loading speed on user satisfaction

- **Independent Variable (IV):** Website loading time (e.g., 1 sec, 3 sec, 5 sec)
- **Dependent Variable (DV):** User satisfaction (rating, feedback)

Key Concept and terminologies

Independent and Dependent Variables

UI Design and User Experience

- **Study Topic:** Effect of UI design on user experience
- **Independent Variable (IV):** Type of UI design (simple vs complex)
- **Dependent Variable (DV):** User experience (ease of use, task completion time)

Key Concept and terminologies

Independent and Dependent Variables

Internet Speed and Video Streaming Quality

Study Topic: Effect of internet speed on video streaming

- **Independent Variable (IV):** Internet speed (Mbps)
- **Dependent Variable (DV):** Video quality (HD, buffering time)

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Key Concept and terminologies

Independent and Dependent Variables

Mobile App Notifications and User Engagement

Study Topic: Effect of notifications on app usage

- **Independent Variable (IV):** Number/frequency of notifications
- **Dependent Variable (DV):** User engagement (time spent, clicks)

Key Concept and terminologies

Extraneous Variable

- An **extraneous variable** is any variable other than the independent variable that may influence the dependent variable.
- These variables are not the main focus but can affect the outcome, leading to inaccurate results if not controlled.

<u>**Example**</u>

- Study: Social media use and personality
  - Extraneous Variable: Time spent exercising
  - Because it affects both personality and social media usage

Key Concept and terminologies

Control:

- **Control** is the process of **minimizing or eliminating the influence of extraneous variables** so that the results of a study are only affected by the independent variable.
- By controlling variables, researchers ensure that only the independent variable affects the dependent variable.
- In any research, many factors (variables) can influence the outcome. However, researchers are usually interested in studying the effect of **one main variable (independent variable)** on another (dependent variable).

Example

- Same classroom, teacher, and time for all students in an experiment

Key Concept and terminologies

If other variables (called **extraneous variables**) are not controlled, they can:

- Distort results
- Create confusion
- Lead to incorrect conclusions

Control helps to:

- Maintain **validity** of the research
- Ensure **reliable results**
- Identify the **true cause-effect relationship**
- Avoid **confusion from unwanted variables**

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Key Concept and terminologies

Study: Effect of teaching method on student performance

- Independent Variable: Teaching method
- Dependent Variable: Student performance

Control Applied:

- Same classroom
- Same teacher
- Same time duration
- Same syllabus

By keeping these factors constant, the researcher ensures that any change in performance is due to the teaching method only.

Key Concept and terminologies

Software Performance

Study: Effect of algorithm on execution time

- Control:
  - Same computer hardware
  - Same input data size
  - Same programming language
- Ensures fair comparison between algorithms.

Key Concept and terminologies

Confounded Relationship between Variable

- A **confounded relationship** occurs when the effect of the independent variable on the dependent variable is mixed with the effect of an extraneous variable.
- The Hidden Factor (Confounding Variable)
- This makes it difficult to determine the true cause of the results.

Example

- Ice cream sales and crime rate increase together
- Confounding variable: Hot weather

Key Concept and terminologies

<u>Example (Confounded Relationship)</u>

- Ice cream sales and crime rate increase at the same time
- It may seem like ice cream causes crime

But the real cause is **hot weather**

- Hot weather increases:
  - Ice cream sales
  - Crime rate (Increase Social Interaction, More chance to conflict)

Key Concept and terminologies

Internet Speed and Website Performance

- **Observation :**Users with faster internet report better website performance
- **Possible Conclusion**
  - Website is well-optimized
- **Confounding Variable**
  - Device performance (CPU, RAM)
- High-end users often have both:
  - Fast internet
  - Powerful devices
- So performance may be due to **device**, not just internet

Key Concept and terminologies

- **Hypothesis :**
- A **hypothesis** is a tentative or testable statement predicting a relationship between variables.
- It provides direction to the research and must be tested before drawing conclusions.

Types

- **Null Hypothesis (H_0):** No relationship exists
- **Alternative Hypothesis (H_1):** A relationship exists

- H_0: Social media has no effect on academic performance
- H_1: Social media affects academic performance

Key Concept and terminologies

Hypothesis (Website Performance Example)

Study Topic: Effect of website loading speed on user experience

Variables

- **Independent Variable (IV):** Website loading speed (fast / slow)
- **Dependent Variable (DV):** User experience (satisfaction, bounce rate, time on site)
- **Null Hypothesis (H_0)**

H_0: Website loading speed has **no effect** on user experience

- **Alternative Hypothesis (H_1)**

H_1: Website loading speed **affects** user experience

Key Concept and terminologies

Experimental and Control Groups

- **Control Group:** A group that is not exposed to the treatment and is kept under normal conditions.
- **Experimental Group:** A group that is exposed to a new condition or treatment.
- **Example:** Study: New teaching method
  - Control Group: Traditional method
  - Experimental Group: New method

Key Concept and terminologies

Treatments :

- **Treatments** are the different conditions or interventions applied to experimental groups.
- They represent the independent variable being tested.

- Study: Fertilizer effect on plant growth
  - Treatment 1: No fertilizer
  - Treatment 2: Organic fertilizer
  - Treatment 3: Chemical fertilizer

Key Concept and terminologies

Website Performance (Loading Speed)

Effect of loading speed on user experience

- **Independent Variable:** Website loading speed
- **Dependent Variable:** User satisfaction / bounce rate

Treatments:

- **Treatment 1:** Fast loading (1 second)
- **Treatment 2:** Moderate loading (3 seconds)
- **Treatment 3:** Slow loading (6 seconds)

Different loading speeds are applied to see how users react.

Types of Research Design

- There are different design types
    - Descriptive
    - Exploratory
    - Experimental
- You can choose the type design based on your objectives.
- High quality research design minimizes bias and strengthens the findings.
- The right research design will help in achieving your research goals.

Types of Research Design:Descriptive

- The descriptive research design involves observing and collecting data on a given topic without attempting to infer cause-and-effect relationships.
- The goal of descriptive research is to provide a comprehensive and accurate picture of the population or phenomenon being studied and to describe the relationships, patterns, and trends that exist within the data.
- Descriptive research methods can include surveys, observational studies, and case studies, and the data collected can be qualitative or quantitative.
- The findings from descriptive research provide valuable insights and inform future research, but do not establish cause-and-effect relationships.

Importance of Descriptive Research in Scientific Studies

<u>Understanding of a Population or Phenomenon</u>

- Descriptive research provides a comprehensive picture of the characteristics and behaviors of a particular population or phenomenon, allowing researchers to gain a deeper understanding of the topic.

Baseline Information

- The information gathered through descriptive research can serve as a baseline for future research and provide a foundation for further studies.

Informative Data

- Descriptive research can provide valuable information and insights into a particular topic, which can inform future research, policy decisions, and programs.

Importance of Descriptive Research in Scientific Studies

Sampling Validation

- Descriptive research can be used to validate sampling methods and to help researchers determine the best approach for their study.

Cost Effective

- Descriptive research is often less expensive and less time-consuming than other research methods, making it a cost-effective way to gather information about a particular population or phenomenon.

Easy to Replicate

- Descriptive research is straightforward to replicate, making it a reliable way to gather and compare information from multiple sources.

Key Characteristics of Descriptive Research Design

Purpose

- The primary purpose of descriptive research is to describe the characteristics, behaviors, and attributes of a particular population or phenomenon.

Participants and Sampling

- Descriptive research studies a particular population or sample that is representative of the larger population being studied. Furthermore, sampling methods can include convenience, stratified, or random sampling.

Data Collection Techniques

- Descriptive research typically involves the collection of both qualitative and quantitative data through methods such as surveys, observational studies, case studies, or focus groups.
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Key Characteristics of Descriptive Research Design

Data Analysis

- Descriptive research data is analyzed to identify patterns, relationships, and trends within the data. Statistical techniques, such as frequency distributions and descriptive statistics, are commonly used to summarize and describe the data.

Focus on Description

- Descriptive research is focused on describing and summarizing the characteristics of a particular population or phenomenon. It does not make causal inferences.

Non-Experimental

- Descriptive research is non-experimental, meaning that the researcher does not manipulate variables or control conditions. The researcher simply observes and collects data on the population or phenomenon being studied.

When Can a Researcher Conduct Descriptive Research?

A researcher can conduct descriptive research in the following situations:

- To better understand a particular population or phenomenon
- To describe the relationships between variables
- To describe patterns and trends
- To validate sampling methods and determine the best approach for a study
- To compare data from multiple sources.

Types of Descriptive Research Design

Survey Research

- Surveys are a type of descriptive research that involves collecting data through self-administered or interviewer-administered questionnaires. Additionally, they can be administered in-person, by mail, or online, and can collect both qualitative and quantitative data.

Observational Research

- Observational research involves observing and collecting data on a particular population or phenomenon without manipulating variables or controlling conditions. It can be conducted in naturalistic settings or controlled laboratory settings.

Types of Descriptive Research Design

Case Study Research

- Case study research is a type of descriptive research that focuses on a single individual, group, or event. It involves collecting detailed information on the subject through a variety of methods, including interviews, observations, and examination of documents.

Focus Group Research

- Focus group research involves bringing together a small group of people to discuss a particular topic or product. Furthermore, the group is usually moderated by a researcher and the discussion is recorded for later analysis.

Types of Descriptive Research Design

Ethnographic Research

- Ethnographic research involves conducting detailed observations of a particular culture or community. It is often used to gain a deep understanding of the beliefs, behaviors, and practices of a particular group.

Advantages of Descriptive Research Design

Provides a Comprehensive Understanding

- Descriptive research provides a comprehensive picture of the characteristics, behaviors, and attributes of a particular population or phenomenon, which can be useful in informing future research and policy decisions.

Non-invasive

- Descriptive research is non-invasive and does not manipulate variables or control conditions, making it a suitable method for sensitive or ethical concerns.

Advantages of Descriptive Research Design

Flexibility

- Descriptive research allows for a wide range of data collection methods, including surveys, observational studies, case studies, and focus groups, making it a flexible and versatile research method.

Cost-effective

- Descriptive research is often less expensive and less time-consuming than other research methods. Moreover, it gives a cost-effective option to many researchers.

Advantages of Descriptive Research Design

Easy to Replicate

- Descriptive research is easy to replicate, making it a reliable way to gather and compare information from multiple sources.

Informs Future Research

- The insights gained from a descriptive research can inform future research and inform policy decisions and programs.

Disadvantages of Descriptive Research Design

Limited Scope

- Descriptive research only provides a snapshot of the current situation and cannot establish cause-and-effect relationships.

Dependence on Existing Data

- Descriptive research relies on existing data, which may not always be comprehensive or accurate.

Lack of Control

- Researchers have no control over the variables in descriptive research, which can limit the conclusions that can be drawn.

Disadvantages of Descriptive Research Design

Bias

- The researcher's own biases and preconceptions can influence the interpretation of the data.

Lack of Generalizability

- Descriptive research findings may not be applicable to other populations or situations.

Lack of Depth

- Descriptive research provides a surface-level understanding of a phenomenon, rather than a deep understanding.

Time-consuming

- Descriptive research often requires a large amount of data collection and analysis, which can be time-consuming and resource-intensive.

What Is Experimental Research Design?

- Experimental research design is a framework of protocols and procedures created to conduct experimental research with a scientific approach using two sets of variables.
- Herein, the first set of variables acts as a constant, used to measure the differences of the second set.
- The best example of experimental research methods is quantitative research.
- Experimental research helps a researcher to gather the necessary data for making better research decisions and determining the facts of a research study.

When Can a Researcher Conduct Experimental Research?

A researcher can conduct experimental research in the following situations :

- When time is an important factor in establishing a relationship between the cause and effect.
- When there is an invariable or never-changing behavior between the cause and effect.
- Finally, when the researcher wishes to understand the importance of the cause and effect.

Importance of Experimental Research Design

- Provides a strong foundation for conducting research and achieving meaningful, publishable results.
- Supports effective decision-making while ensuring the research is well-structured for easy data collection and analysis.
- Helps clearly define the research question, scope, and boundaries of the study.
- Encourages proper planning and organization before conducting experiments.
- Enhances the reliability and accuracy of results while reducing the risk of inconclusive outcomes.

Types of Experimental Research Designs

Pre-experimental Research Design

- A research study could conduct pre-experimental research design when a group or many groups are under observation after implementing factors of cause and effect of the research.
- The pre-experimental design will help researchers understand whether further investigation is necessary for the groups under observation.

<u>Pre-experimental research is of three types :</u>

- One-shot Case Study Research Design
- One-group Pretest-posttest Research Design
- Static-group Comparison

Types of Experimental Research Designs

- **One-Shot Case Study Research Design**
  - A single group is selected and exposed to a treatment or experiment.
  - No pretest is conducted before applying the treatment.
  - Observation is made only after the treatment.
  - Lacks comparison and control, so results may not be highly reliable.

Types of Experimental Research Designs

- **One-Group Pretest-Posttest Research Design**
  - A single group is observed before and after the treatment.
  - A pretest is conducted to measure the initial condition.
  - The treatment is then applied.
  - A posttest is conducted to measure any changes.
  - Helps identify changes, but still lacks a control group for strong conclusions.

Types of Experimental Research Designs

- **Static-Group Comparison Design**
  - Two groups are used: one experimental group and one comparison group.
  - The experimental group receives the treatment, while the other does not.
  - No random assignment of groups is done.
  - Only posttest observations are made for both groups.
  - Provides some comparison but lacks full control over external variables.

Types of Experimental Research Designs

True Experimental Research Design

- A true experimental research design relies on statistical analysis to prove or disprove a researcher's hypothesis.
- It is one of the most accurate forms of research because it provides specific scientific evidence.
- Furthermore, out of all the types of experimental designs, only a true experimental design can establish a cause-effect relationship within a group.

Types of Experimental Research Designs

True experiment, a researcher must satisfy these three factors :

- There is a control group that is not subjected to changes and an experimental group that will experience the changed variables
- A variable that can be manipulated by the researcher
- Random distribution of the variables

This type of experimental research is commonly observed in the physical sciences.

Types of Experimental Research Designs

Quasi-experimental Research Design

- The word “Quasi” means similarity.
- A quasi-experimental design is similar to a true experimental design.
- However, the difference between the two is the assignment of the control group.
- In this research design, an independent variable is manipulated, but the participants of a group are not randomly assigned.
- This type of research design is used in field settings where random assignment is either irrelevant or not required.
- The classification of the research subjects, conditions, or groups determines the type of research design to be used.

Advantages of Experimental Research

- Researchers have firm control over variables to obtain results.
- The subject does not impact the effectiveness of experimental research. Anyone can implement it for research purposes.
- The results are specific.
- Post results analysis, research findings from the same dataset can be repurposed for similar research ideas.
- Researchers can identify the cause and effect of the hypothesis and further analyze this relationship to determine in-depth ideas.
- Experimental research makes an ideal starting point. The collected data could be used as a foundation to build new research ideas for further studies.

Example of Experimental Research

- The researcher begins by collecting plant samples for the study.
- The samples are randomly divided into two groups to avoid bias.
- One group is exposed to sunlight to allow photosynthesis.
- The other group is kept in a dark environment without sunlight.
- All other variables (water, nutrients, soil, etc.) are kept constant to ensure fairness.
- After a set period, both groups are tested using biochemical methods.
- The results of both groups are compared carefully.
- Any differences observed in the plants are attributed to sunlight.
- This helps confirm a cause-and-effect relationship between sunlight and plant changes.

Example of Experimental Research

Topic: Effect of Caching Mechanism on Web Application Performance

- A researcher selects a web application as the subject of the experiment.
- The application is deployed in a controlled environment (same server, network conditions, and database).
- Two experimental setups are created:
  - **Group A:** Web application without caching
  - **Group B:** Web application with caching enabled (e.g., server-side or browser caching)

Example of Experimental Research

- All other variables (hardware, traffic load, database queries, and network speed) are kept constant.
- A fixed number of user requests (simulated traffic) are sent to both setups.
- Performance metrics such as page load time, response time, and server CPU usage are recorded.
- The results from both groups are collected and compared.
- Any improvement in performance is attributed to the caching mechanism.
- The experiment helps establish a cause-and-effect relationship between caching and system performance.
- **Conclusion:**
  This experimental design shows how a specific factor (caching) impacts system performance while controlling other variables.

Exploratory Research Design

- Exploratory research design is a type of research conducted when the problem is not clearly defined or when the researcher wants to gain deeper understanding and new insights into a phenomenon.
- Its main purpose is to explore ideas, identify patterns, formulate hypotheses, and clarify concepts for future detailed investigation.
- exploratory research aims "*to gain familiarity with a phenomenon or to achieve new insights into it*."
- Exploratory research design is a flexible and unstructured approach used to investigate problems that are not well understood.
- It helps researchers discover ideas, generate hypotheses, and identify variables for further study.

Objectives of Exploratory Research Design

- To gain familiarity with a topic or problem
- To identify research variables and relationships
- To generate new ideas and insights
- To formulate hypotheses for future studies
- To clarify ambiguous concepts and problems

Characteristics of Exploratory Research Design

- Flexible and adaptable in nature
- Does not follow rigid procedures
- Mostly qualitative in approach
- Conducted when little prior information is available
- Helps in problem formulation rather than problem solving
- Uses small samples and informal methods

Steps in Exploratory Research

- Identify the research problem
- Conduct preliminary investigation
- Review related literature
- Collect qualitative or secondary data
- Analyze information and identify patterns
- Develop hypotheses or directions for future research

Advantages of Exploratory Research

- Helps understand unfamiliar problems
- Generates new ideas and hypotheses
- Flexible and inexpensive
- Useful as a foundation for further research
- Assists in identifying important variables

Limitations of Exploratory Research

- Findings are not conclusive
- Results cannot usually be generalized
- High possibility of researcher bias
- Lack of statistical precision
- May not provide definite answers

Methods of Exploratory Research

- Exploratory research design is conducted when the researcher has limited knowledge about the problem and wishes to gain preliminary understanding, discover ideas, identify variables, and formulate hypotheses for future investigation.
- Since exploratory research is flexible and open-ended in nature, several methods are commonly used for collecting information and obtaining insights into the research problem.

Methods of Exploratory Research

The important methods used in exploratory research design are:

- Literature Survey
- Experience Survey
- Case Study Method
- Focus Group Discussion
- Pilot Study

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Methods of Exploratory Research

Literature Survey

- The literature survey, also known as literature review, is one of the most important methods used in exploratory research.
- It involves a systematic study of existing literature related to the research problem.
- The researcher examines books, journals, research papers, theses, reports, conference proceedings, government publications, and online resources to gather background information about the topic.
- Research methodology emphasize literature review as an essential component in understanding the research problem and identifying research gaps.

Methods of Exploratory Research

Experience Survey

- Experience survey is a method in which information is collected from persons who possess special knowledge, experience, or expertise related to the research problem.
- Such individuals may include experts, managers, researchers, professionals, customers, or experienced employees.
- The method is particularly useful when very little information is available regarding the problem under investigation.

Methods of Exploratory Research

Case Study Method

- The case study method involves an intensive and detailed investigation of a single unit such as an individual, organization, institution, event, or community.
- The objective is to understand the problem thoroughly in its real-life context.
- This method is widely used in social sciences, business studies, psychology, education, and management research.

Methods of Exploratory Research

Focus Group Discussion

- Focus Group Discussion (FGD) is a qualitative research technique in which a small group of individuals discusses a specific topic under the guidance of a moderator. The method helps the researcher understand opinions, attitudes, beliefs, perceptions, and experiences of participants.

Composition of Focus Group

A focus group generally consists of:

- 6 to 12 participants
- Individuals having similar characteristics or experiences
- One trained moderator

Methods of Exploratory Research

Pilot Study

- A pilot study is a small-scale preliminary investigation conducted before the main research study.
- It is carried out to test the feasibility, reliability, validity, and practicality of the research design and research instruments.
- The pilot study helps the researcher identify and correct possible problems before conducting the final investigation.

Example of Exploratory Research

- A company notices a sudden decline in product sales but does not know the reason.
- It conducts exploratory research through customer interviews and focus groups to identify possible causes such as pricing, competition, or customer preferences.

Example of Exploratory Research

Topic: Declining Sales of a Smartphone Brand

Introduction

- Suppose a smartphone company notices that its sales have suddenly decreased over the last six months. The management does not clearly know the exact reasons for the decline. Since the problem is not properly understood, the company decides to conduct an exploratory research study to identify possible causes and obtain preliminary insights.
- In this situation, exploratory research design is most suitable because it helps the researcher **explore the problem**, **gather ideas**, and identify important **factors affecting sales**.

Example of Exploratory Research

Problem Statement

- The company is experiencing a continuous decline in smartphone sales, but the exact causes are unknown.

Objectives of the Study

The exploratory research aims to:

- Identify possible reasons for declining sales
- Understand customer perceptions about the brand
- Explore competitor influence
- Identify product-related issues
- Generate hypotheses for future detailed research

Example of Exploratory Research

Methods Used in the Exploratory Research

- The researcher uses different exploratory research methods to investigate the problem.

Literature Survey

The company first reviews:

- Previous sales reports
- Market research reports
- Customer complaints
- Industry publications
- Research articles on consumer buying behavior

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Example of Exploratory Research

Findings from Literature Survey

The review suggests that smartphone sales may decline because of:

- Poor battery performance
- High pricing
- Strong competition
- Lack of innovative features
- Negative online reviews

Thus, the literature survey provides initial understanding of possible factors affecting sales.

Example of Exploratory Research

Experience Survey

The company then conducts interviews with:

- Sales managers
- Retail shop owners
- Marketing experts
- Existing customers
- Technical support staff

Findings from Experience Survey

Experts suggest that:

- Customers prefer competitors with better camera quality
- The company's after-sales service is weak
- Younger consumers prefer trendy designs
- Customers consider the price too high

These expert opinions help identify practical business problems.

Example of Exploratory Research

Case Study Method

The company performs a detailed case study of:

- One successful competitor brand
- One retail store with declining sales

Findings from Case Study

The study reveals:

- Competitors frequently launch updated models
- Competitors provide better customer support
- Online marketing strategies are more effective

This detailed investigation provides deeper understanding of market conditions.

Example of Exploratory Research

- **Focus Group Discussion**

The company organizes focus group discussions with:

- College students
- Young professionals
- Existing smartphone users
- Each group contains 8–10 participants.

Discussion Topics

- Preferred smartphone features
- Brand image
- Price expectations
- Problems with current products

Findings from Focus Group

Participants express that:

- Battery backup is poor
- Camera quality is outdated
- The design is not attractive
- Advertisements are not appealing

The discussion reveals customer attitudes and expectations.

Example of Exploratory Research

Pilot Study

- Before conducting a large customer survey, the company tests its questionnaire on 30 customers.

Purpose of Pilot Study

The pilot study helps:

- Identify unclear questions
- Improve questionnaire structure
- Estimate response time
- Check reliability of questions

After modifications, the company prepares the final survey for large-scale research.

Example of Exploratory Research

Outcome of Exploratory Research

After conducting exploratory research, the company identifies several possible causes of declining sales:

- High product price
- Weak after-sales service
- Outdated product features
- Poor advertising strategy
- Strong market competition

The exploratory study does not provide final conclusions but helps the company clearly define the problem and formulate hypotheses for further descriptive or causal research.

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Example of Exploratory Research

Possible Hypotheses Developed

Based on exploratory findings, the company may formulate hypotheses such as:

- High product prices reduce customer purchase intention.
- Better camera quality increases customer preference.
- Effective after-sales service improves customer satisfaction.

Measuring the Human in Research

- In research, particularly in social sciences, psychology, management, education, healthcare, and Human-Computer Interaction (HCI), researchers often need to measure *human behavior*, *attitudes*, *emotions*, *intelligence*, *perceptions*, and *performance*. This process is known as **measuring the human in research**.
- Human characteristics are abstract and cannot usually be measured directly like physical quantities such as *length* or *weight*. Therefore, researchers use scientific tools, scales, and techniques to convert human responses and behaviors into measurable data.
- Research methodology describe measurement as an important aspect of scientific investigation because accurate measurement improves the reliability and validity of research findings.

Measuring the Human in Research

Importance of Measuring Humans in Research

Measuring humans is important because it enables researchers to:

- Study human behavior scientifically
- Compare individuals or groups
- Analyze relationships among variables
- Evaluate performance and effectiveness
- Make objective decisions
- Develop theories and predictions
- Improve systems, products, and services

For example:
- A psychologist measures anxiety levels.
- A teacher measures learning outcomes.
- A software engineer measures user satisfaction.

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Measuring the Human in Research

Characteristics of Human Measurement

Human measurement differs from physical measurement because human behavior is:

- Abstract
- Dynamic
- Subjective
- Influenced by emotions and environment
- Difficult to observe directly

Therefore, specialized tools and measurement techniques are required.

Measuring the Human in Research

Areas of Human Measurement

Area	Examples
Cognitive Factors	Intelligence, memory, learning ability
Emotional Factors	Stress, anxiety, happiness
Behavioral Factors	Habits, interaction patterns
Social Factors	Teamwork, communication
Performance Factors	Productivity, task efficiency
Attitudinal Factors	Opinions, beliefs, satisfaction

Methods Used for Measuring Humans in Research

Observation Method

- Observation involves systematically watching and recording human behavior in natural or controlled environments.
- Research methodology describe observation as an important method of collecting primary data.

Types of Observation

  - **Participant Observation:** The researcher participates in the activity being observed.
  - **Non-Participant Observation :** The researcher only observes without participating.

Methods Used for Measuring Humans in Research

- **Example:** Observing how users interact with a mobile application interface.

Advantages

- Captures actual behavior
- Useful for behavioral studies
- Does not rely only on self-reporting

Limitations

- Observer bias may occur
- Internal feelings and emotions cannot always be observed

Methods Used for Measuring Humans in Research

Questionnaire Method

- A questionnaire is a structured set of questions used to collect information from participants.
- Research methodology texts discuss questionnaires as one of the major tools for collecting primary data.

Types of Questions

Closed-Ended Questions: Participants select answers from predefined options.

Open-Ended Questions: Participants express opinions freely.

Methods Used for Measuring Humans in Research

- **Example:** A survey measuring student satisfaction with online learning systems.

Advantages

- Suitable for large populations
- Economical and easy to administer
- Data can be statistically analyzed

Limitations

- Participants may give inaccurate responses
- Misunderstanding of questions is possible

Methods Used for Measuring Humans in Research

Interview Method

- Interview is a method in which researchers collect information through direct interaction with participants.

Types of Interviews

- **Structured Interview:** Uses predetermined questions.
- **Unstructured Interview:** Flexible discussion without fixed sequence.
- **Semi-Structured Interview:** Combines both structured and flexible approaches.

Methods Used for Measuring Humans in Research

Example: Interviewing software developers about workplace stress and productivity.

Advantages

- Provides detailed information
- Clarification is possible
- Allows exploration of emotions and opinions

Limitations

- Time-consuming
- Interviewer bias may influence responses

Methods Used for Measuring Humans in Research

Psychological Tests

- Psychological tests are standardized instruments used to measure mental abilities, personality traits, emotions, and psychological conditions.

Examples

- Intelligence Quotient (IQ) tests
- Personality tests
- Aptitude tests
- Anxiety and depression scales

Methods Used for Measuring Humans in Research

Advantages

- Scientifically standardized
- Reliable and valid

Limitations

- Requires expert administration
- Cultural and language bias may occur

Methods Used for Measuring Humans in Research

Rating Scales

- Rating scales are tools used to measure attitudes, opinions, or perceptions by assigning numerical or descriptive values.
- Scaling techniques as important measurement tools in human research.

Methods Used for Measuring Humans in Research

<u>Types of Rating Scales</u>

Likert Scale

- Measures degree of agreement or disagreement.

<u>Example</u>:

- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

Methods Used for Measuring Humans in Research

Semantic Differential Scale

- Measures attitudes between opposite adjectives.
- Example:
  Efficient ___ ___ ___ Inefficient

Methods Used for Measuring Humans in Research

Applications

- User satisfaction measurement
- Product usability studies
- Employee feedback systems

Advantages

- Easy to use
- Suitable for quantitative analysis

Limitations

- Responses may be subjective
- Central tendency bias may occur

Methods Used for Measuring Humans in Research

Biometric and Physiological Measurement

- Physiological measurement involves recording biological responses using scientific instruments and sensors.

Examples

- Heart rate monitoring
- Eye tracking
- EEG (brain activity measurement)
- Facial emotion detection
- Skin conductance measurement

Methods Used for Measuring Humans in Research

Measurement Scales Used in Human Research

- Research methodology literature identifies **four** major scales of measurement.

Nominal Scale

- Used for classification only.

Example

- Gender:
  - Male
  - Female

Methods Used for Measuring Humans in Research

Ordinal Scale

- Ranks data according to order.

Example

Performance levels:

- High
- Medium
- Low

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Methods Used for Measuring Humans in Research

Interval Scale

- Measures numerical differences without absolute zero.

Example: Temperature scale

Ratio Scale

- Contains an absolute zero point.

Example: Reaction time in milliseconds

Reliability and Validity in Human Measurement

Reliability

- Reliability refers to the consistency and stability of measurement results.
- A reliable instrument produces similar results under similar conditions.
- Research methodology books explain reliability as a major criterion for good measurement.

Validity

- Validity refers to the accuracy of the measurement instrument.
- A valid instrument measures what it is intended to measure.
- Research methodology literature identifies several forms of validity including content validity, construct validity, and criterion validity.

Reliability and Validity in Human Measurement

Challenges in Measuring Humans

Measuring human characteristics is difficult because:

- Human behavior changes over time
- Emotions influence responses
- Cultural differences affect interpretation
- Participants may provide false information
- Observer bias may occur

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Preparing Research Design: Universe in Research

- Research design is the blueprint of a research study.
- It provides a systematic framework for collecting, measuring, and analyzing data.
- A well-prepared research design helps the researcher achieve reliable, valid, and objective findings.
- <u>Important components of research design include the</u>
    - identification of the universe and population
    - determination of sample size
    - selection of samples
    - selection of research instruments.

Preparing Research Design: Universe in Research

- In research methodology, the term ***universe*** refers to the entire group of individuals, objects, events, or elements possessing common characteristics that are relevant to a particular study.
- It represents the total collection of units from which information is required.
- In research methodology, the **universe** refers to the complete collection of units, objects, individuals, or events about which the researcher wants to obtain information.
- The universe is also called the **aggregate** or **population universe**.
- Universes are classified into different types based on their size and characteristics.
- Type of universe is important because it helps researchers choose appropriate sampling methods, research designs, and statistical tools.

Preparing Research Design: <u>Universe in Research</u>

The universe may be:

- Finite or infinite
- Real or hypothetical
- Homogeneous or heterogeneous

Examples

- All university students in Nepal.
- All software engineers working in Kathmandu.
- All users of a mobile banking application.
- All social media posts related to a political event.

Preparing Research Design: Universe in Research

Finite Universe

- A **finite universe** is a universe that contains a fixed, limited, and countable number of elements.
- The total number of units in the universe can be determined exactly.
- In a finite universe:
  - The size of the universe is known.
  - Every element can theoretically be identified and listed.
  - Sampling becomes easier and more accurate.

Examples

- Students enrolled in a college.
- Employees in a software company.
- Registered voters in a municipality.
- Computers installed in a laboratory.
- Users subscribed to a mobile application.

Preparing Research Design: <u>Universe in Research</u>

Research Topic

“User Satisfaction with a College ERP System”

Universe

- All students of Patan Multiple College.
- Suppose the college has: 2,500 students
- Since the exact number is known, the universe is finite.

Preparing Research Design: Universe in Research

Infinite Universe

- An **infinite universe** is a universe in which the number of elements is unlimited, uncountable, or impossible to determine exactly.
- In such cases:
  - New elements may continuously emerge.
  - The total size cannot be accurately measured.
  - Researchers usually depend on probability theory and estimation.

Examples

- Online transactions occurring globally.
- Internet searches performed every second.
- Tweets posted worldwide.
- Stars in the universe.
- Continuous production defects in manufacturing.

Preparing Research Design: Universe in Research

Research Topic : “Analysis of Global Social Media Trends”

Universe

- All tweets posted on Twitter worldwide.
- Since tweets are continuously generated every second, the universe is practically infinite.

Research Topic : “Cybersecurity Threat Detection”

- **Universe**
- All network packets transmitted over the internet.
- The data flow is continuous and practically endless.

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Preparing Research Design: <u>Universe in Research</u>

Homogeneous Universe

- A **homogeneous universe** is a universe in which all units possess similar or nearly identical characteristics relevant to the study.
- The elements are ***<u>uniform</u>*** with respect to:
  - Age
  - Qualification
  - Profession
  - Income level
  - Technical background
  - Other study variables
  - Because of similarity among units, sampling becomes simpler.

Preparing Research Design: Universe in Research

Examples

- Employees working in the same department.
- Students of the same class.
- Machines manufactured using identical specifications.
- Servers running the same operating system.
- Users with similar technical skills.

Preparing Research Design: <u>Universe in Research</u>

Research Topic: “Performance Analysis of Python Programmers”

Universe

- Final-year computer science students trained in Python programming.
- Since all students have similar educational backgrounds and programming exposure, the universe is homogeneous.

Research Topic: “Evaluation of Cloud Computing Knowledge”

Universe

- Cloud engineers working in the same company.
- Because their training and work profiles are similar, the universe is homogeneous.

Preparing Research Design: Universe in Research

Heterogeneous Universe

- A **heterogeneous universe** is a universe in which the units differ significantly in their characteristics.
- The elements **may vary** in:
  - Age
  - Gender
  - Occupation
  - Education
  - Economic background
  - Technical knowledge
  - Geographical location

  Because of diversity, researchers must use more careful sampling methods.

Preparing Research Design: Universe in Research

Examples

- Citizens of an entire country.
- Internet users worldwide.
- Consumers of an online shopping platform.
- Hospital patients from different age groups.
- Social media users from different countries.

Preparing Research Design: Universe in Research

Research Topic : "Digital Literacy among Internet Users"

Universe

- All internet users in Nepal.
- This universe is heterogeneous because users differ in:
    - Age
    - Education
    - Technical knowledge
    - Location
    - Language
    - Internet access

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Preparing Research Design: Universe in Research

Research Topic: “Adoption of Artificial Intelligence Tools”

Universe

- Professionals from healthcare, education, banking, and IT sectors.
- These groups have different technical backgrounds and experiences.

Preparing Research Design: Universe in Research

Importance of Understanding Universe Types

Understanding the type of universe helps researchers:

- Select appropriate sampling methods.
- Determine sample size accurately.
- Choose proper statistical tools.
- Improve validity and reliability.
- Reduce research bias and errors.

For example:

- Homogeneous universes often use smaller random samples.
- Heterogeneous universes may require stratified sampling.
- Infinite universes rely heavily on probabilistic estimation.

Preparing Research Design: <u>Population in Research</u>

- Population refers to the complete set of individuals or elements that satisfy the sampling criteria established by the researcher. It is a specific portion of the universe selected for investigation.
- Population is also called the *target population*.
- In research methodology, the term *population* refers to the complete set of individuals, objects, events, organizations, observations, or elements that satisfy specific criteria relevant to a research study.
- The population forms the central focus of investigation because researchers aim to draw conclusions about it through systematic data collection and analysis.

Preparing Research Design: Population in Research

Example

If the universe is all internet users in Nepal, the population may be:

- College students using social media applications.
- E-commerce users in Kathmandu.

Characteristics of Population

- Clearly identifiable
- Relevant to the research problem
- Measurable and accessible
- May be large or small

Preparing Research Design: Population in Research

A population may consist of:

- people,
- institutions,
- software systems,
- documents,
- transactions,
- machines,
- or any measurable entities depending on the nature of the research.

Since studying the entire population is often difficult, researchers select samples from the population. Therefore, proper identification of the population is essential for obtaining reliable and valid research results.

Preparing Research Design: <u>Population in Research</u>

In research methodology, populations are commonly classified into:

- Target Population
- Accessible Population
- Study Population

Understanding these types helps researchers:

- define research boundaries,
- choose suitable sampling techniques,
- reduce bias,
- improve data collection,
- and increase the validity of research findings.

Preparing Research Design: Population in Research

Target Population

- The **target population** refers to the entire group of individuals or elements about which the researcher wants to draw conclusions.
- It represents the complete set of units possessing the characteristics relevant to the research problem. The researcher’s findings, interpretations, and generalizations are intended to apply to this population.
- The target population is broader in scope and may not always be directly accessible for data collection.

Preparing Research Design: Population in Research

Research Topic : “Impact of Artificial Intelligence Chatbots on Learning”

Target Population

- All university students using AI-based learning systems.
- The researcher aims to generalize the findings to all such students, regardless of location or institution.

Research Topic : “User Satisfaction with Mobile Banking Applications”

Target Population

- All mobile banking users in Nepal.
- The research findings are intended to represent the experiences of all users.

Preparing Research Design: Population in Research

Importance of Target Population

The target population:

- defines the scope of the research
- guides sample selection
- helps identify research variables
- supports statistical generalization
- and improves research clarity

Preparing Research Design: <u>Population in Research</u>

Accessible Population

- The **accessible population** refers to the portion of the target population that is practically available to the researcher for data collection.
- Although the researcher may wish to study the entire target population, practical limitations such as:
- time, cost, geographical barriers, permissions, technical limitations, and accessibility which make it impossible to reach every member.

Therefore, researchers focus on the accessible population.

Preparing Research Design: Population in Research

Research Topic: “Effectiveness of Online Learning Platforms”

Target Population

- All university students using online learning systems in Nepal.

Accessible Population

- Students from universities located in Kathmandu Valley.

Because the researcher can practically access only these universities, they form the accessible population.

Preparing Research Design: Population in Research

Research Topic : “Cybersecurity Awareness among Employees”

Target Population

- Employees working in IT companies throughout Nepal.

Accessible Population

- Employees from five IT companies willing to participate.

Preparing Research Design: Population in Research

Importance of Accessible Population

- Accessible population:
- makes research practically feasible,
- reduces cost and time,
- improves manageability,
- and supports efficient data collection.

Preparing Research Design: <u>Population in Research</u>

Study Population

- The **study population** refers to the final group of individuals or elements selected for actual investigation in the research study.
- It is the population from which the sample is directly drawn.
- The study population satisfies:
  - the research criteria
  - inclusion conditions,
  - and practical considerations established by the researcher.

The study population is usually derived from the accessible population.

Preparing Research Design: Population in Research

Importance of Study Population

Study population:

- provides actual research data
- ensures relevance
- improves research precision
- and supports valid conclusions

Preparing Research Design: Population in Research

Research Topic: “Analysis of Mobile App Security Awareness”

Target Population

- All smartphone users in Nepal.

Accessible Population

- Smartphone users in Pokhara Metropolitan City.

Study Population

- University students using Android banking applications.

Preparing Research Design: Sample in Research

Sample and Sample Size

- A sample is a subset of the population selected for detailed investigation.
- Since studying the entire population is often difficult, costly, and time-consuming, researchers collect data from a representative sample.

Need for Sampling

- Saves time
- Reduces cost
- Makes data collection manageable
- Enables faster analysis
- Suitable for large populations

Preparing Research Design: Sample in Research

Characteristics of a Good Sample

A good sample should be:

- Representative
- Adequate in size
- Free from bias
- Scientifically selected
- Reliable and valid

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Preparing Research Design: Sample in Research

- Sample size refers to the number of units or respondents included in a research study.

Importance of Sample Size

- Influences reliability of findings
- Affects accuracy of statistical analysis
- Determines generalizability of results
- Reduces sampling error

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Preparing Research Design: Sample in Research

Nature of Research

- Exploratory studies generally require smaller samples, while descriptive and experimental studies require larger samples.

Population Size

- Larger populations usually require larger samples.

Research Objectives

- Complex objectives demand larger and more representative samples.

Availability of Resources

- Time, cost, manpower, and technical facilities affect sample size.

Desired Accuracy

- Higher accuracy requires larger samples.

Variability in Population

- Highly heterogeneous populations require larger samples.

Preparing Research Design: Sample in Research

Methods for Determining Sample Size

An appropriate sample size is essential because:

- it improves accuracy of findings,
- reduces sampling error,
- enhances reliability and validity,
- supports statistical analysis,
- and helps in proper generalization of results.

Preparing Research Design: Sample in Research

Researchers use different methods for determining sample size depending on:

- the nature of research,
- population size,
- required accuracy,
- available resources,
- and statistical requirements.

The major methods for determining sample size are:

- Census Method
- Statistical Formula Method
- Rule of Thumb Method
- Software-Based Determination Method

Preparing Research Design: Sample in Research

Census Method

The **Census Method** is a method in which data are collected from every element or unit of the entire population. Instead of selecting a sample, the researcher studies the complete population.

In this method:

- every individual is included,
- no sampling is performed,
- and the results represent the population directly.

The census method is also called:

- complete enumeration method,
- total population study,
- or complete survey method.

Preparing Research Design: <u>Sample in Research</u>

Characteristics of Census Method

1. Entire Population is Studied

- No sampling is involved.

2. High Accuracy

- Since all units are investigated, sampling error is eliminated.

3. Comprehensive Information

- Detailed and complete data are obtained.

4. Resource Intensive

- Requires substantial time, cost, and manpower.

5. Suitable for Small Populations

- Most useful when population size is manageable

Preparing Research Design: Sample in Research

Suitable Conditions for Census Method

The census method is appropriate when:

1. Population Size is Small
2. High Accuracy is Required
3. Population is Easily Accessible
4. Sufficient Resources are Available
5. Government or Institutional Studies

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Preparing Research Design: Sample in Research

Applications of Census Method

The census method is widely used in:

- national population censuses,
- institutional surveys,
- educational assessments,
- inventory analysis,
- small organizational studies,
- and laboratory research.

Preparing Research Design: <u>Sample in Research</u>

Statistical Formula Method

- The **Statistical Formula Method** is a scientific approach used to determine sample size through mathematical and statistical formulas.
- Researchers use statistical formulas to calculate the minimum number of respondents required for reliable results.

<u>This method is widely used in:</u>

- quantitative research,
- survey research,
- social sciences,
- healthcare studies,
- business research,
- and computer science investigations.

Preparing Research Design: Sample in Research

Common Formula

One of the most widely used formulas is:

$$n = \frac{Z^2 pq}{e^2}$$

Where:

- n = required sample size
- Z = standard normal value
- p = estimated population proportion
- $q = 1 - p$
- e = acceptable sampling error

Preparing Research Design: Sample in Research

Sample Size (n)

- Represents the number of respondents required for the study.

Standard Normal Value (Z)

- The Z value depends on the confidence level selected by the researcher.
- Higher confidence levels require larger sample sizes.

Preparing Research Design: <u>Sample in Research</u>

Estimated Proportion (p)

- Represents the proportion of the population having a particular characteristic.
- If no prior information exists, researchers commonly use:

$$p = 0.5$$

- because it gives the maximum sample size.

Preparing Research Design: Sample in Research

q Value

- $q = 1 - p$

If:

$$p = 0.5$$

then:

$$q = 0.5$$

Preparing Research Design: Sample in Research

Acceptable Error (e)

- Represents the margin of error acceptable in the study.

Common values:

- 5% = 0.05
- 3% = 0.03
- 1% = 0.01

Smaller error margins require larger samples.

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Preparing Research Design: Sample in Research

To calculate the Z-score by hand, use this formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- Z = The standard normal value (Z-score)
- X = The specific data point or value you are examining
- μ (mu) = The mean (average) of the population
- σ (sigma) = The standard deviation of the population

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Preparing Research Design: <u>Sample in Research</u>

- Your score (X) = 85
- The class average (μ) = 70
- The standard deviation (σ) = 10

$$Z = \frac{85 - 70}{10}$$

$$Z = \frac{15}{10}$$

$$Z = 1.5$$

Your Z-score is **1.5**. This means your test score was 1.5 standard deviations *above* the class average.

Preparing Research Design: <u>Sample in Research</u>

Research Topic

"Student Satisfaction with Online Learning Systems"

Suppose:

- Confidence level = 95%
- $Z = 1.96$
- $p = 0.5$
- $q = 0.5$
- $e = 0.05$

Using the formula:

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2}$$

Above calculation gives the required sample size is approximately **384 respondents**.

Preparing Research Design: Sample in Research

Advantages of Statistical Formula Method

- **Scientifically Accurate:** Provides statistically reliable sample sizes.
- **Reduces Sampling Error:** Improves precision.
- **Widely Accepted:** Commonly used in academic and professional research.
- **Suitable for Large Populations:** Useful when census studies are impossible.
- **Supports Generalization:** Improves validity of conclusions.

Preparing Research Design: Sample in Research

Disadvantages of Statistical Formula Method

- **Requires Statistical Knowledge:** Researchers must understand formulas and assumptions.
- **Depends on Correct Estimation:** Wrong assumptions affect sample accuracy.
- **May Require Large Samples:** High confidence levels increase sample size.
- **Complex for Beginners:** Mathematical calculations may be difficult.

Preparing Research Design: Sample in Research

Applications of Statistical Formula Method

Used extensively in:

- social science surveys,
- healthcare research,
- educational studies,
- AI and machine learning research,
- cybersecurity studies,
- and market analysis.

Preparing Research Design: Sample in Research

Rule of Thumb Method

The Rule of Thumb Method determines sample size based on:

- practical experience,
- previous research,
- expert opinion,
- or commonly accepted guidelines.

Instead of using strict statistical formulas, researchers select sample sizes using general recommendations.

- This method is commonly used in exploratory and pilot studies.

Preparing Research Design: <u>Sample in Research</u>

Characteristics of Rule of Thumb Method

- **Simple and Practical:** No complex calculations required.
- **Experience-Based:** Relies on prior studies and researcher judgment.
- **Flexible :**Researchers can adjust sample sizes according to circumstances.
- **Less Scientific:** May not provide exact statistical precision.

Preparing Research Design: <u>Sample in Research</u>

Common Rules of Thumb

- **Minimum Sample Rule**
  - at least 30 respondents.
- **Percentage Rule**
  - A fixed percentage of the population may be selected.
- **Multivariate Analysis Rule**
  - 5 to 10 respondents per variable.
- **Experimental Research Rule**
  - Minimum 15–30 participants per group.

Preparing Research Design: Sample in Research

Advantages of Rule of Thumb Method

- **Easy to Use:** No statistical expertise required.
- **Saves Time:** Quick sample estimation possible.
- **Suitable for Pilot Studies:** Useful in exploratory investigations.
- **Practical in Resource Constraints:** Helpful when resources are limited.

Preparing Research Design: <u>Sample in Research</u>

Disadvantages of Rule of Thumb Method

- **Less Accurate:** May not ensure statistical validity.
- **Subjective:** Depends on researcher judgment.
- **Limited Scientific Basis:** Results may lack precision.
- **Risk of Inadequate Samples:** Poor estimates may reduce reliability.

Preparing Research Design: Sample in Research

Applications of Rule of Thumb Method

- pilot studies,
- exploratory research,
- qualitative research,
- classroom projects,
- and preliminary investigations.

Preparing Research Design: Sample in Research

Software-Based Determination Method

- In modern research, specialized software tools are used to determine sample size scientifically and efficiently.

These software programs perform:

- statistical calculations,
- power analysis,
- confidence interval estimation,
- and sampling optimization automatically.

Preparing Research Design: Sample in Research

Common Software Tools

- **SPSS:** Widely used for statistical analysis and sample estimation.
- **G*Power:** Popular for calculating sample size and statistical power.
- **R Programming:** Provides advanced statistical packages for sample determination.
- **Online Sample Size Calculators:** Web-based tools for quick calculations.
- **SAS(**Statistical Analysis System) **and STATA:** Professional statistical software packages.

Preparing Research Design: Sample in Research

Characteristics of Software-Based Determination

- **Automated Calculations:** Reduces manual errors.
- **High Accuracy:** Provides scientifically valid estimates.
- **Advanced Statistical Support:** Suitable for complex research designs.
- **Time Efficient:** Quickly generates sample recommendations.

Preparing Research Design: Sample in Research

Advantages of Software-Based Determination

- **Highly Accurate:** Provides statistically optimized sample sizes.
- **Saves Time:** Automatic calculations are fast.
- **Supports Advanced Research:** Useful for complex models and experiments.
- **Reduces Human Error:** Improves reliability.
- **Professional Standard:** Widely accepted in academic research.

Disadvantages of Software-Based Determination

- **Requires Technical Knowledge:** Researchers must understand software operation.
- **Software Costs:** Some tools are expensive.
- **Dependence on Input Accuracy:** Incorrect assumptions affect results.
- **Learning Curve:** Beginners may face difficulties.

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Preparing Research Design: Sample in Research

Applications of Software-Based Determination

Used extensively in:

- AI and machine learning,
- healthcare research,
- experimental studies,
- business analytics,
- engineering research,
- and quantitative social sciences.

Preparing Research Design: Sample Selection

- **Sample selection** is the process of choosing a group of individuals or units from a larger population for research.
- It helps researchers study a **smaller** group **instead** of the **entire** population.
- The selected sample should represent the **characteristics** of the **population**.
- Proper sample selection **improves** the **accuracy** and **reliability** of research findings.
- It helps **save** time, cost, and effort in data collection.
- An **appropriate** sampling method **reduces** the chances of **bias** and **errors**.
- The results obtained from the sample can be used to draw conclusions about the whole population.

Preparing Research Design: <u>Sampling Techniques</u>

- Sampling techniques are methods used to select a sample from a population.
- They help researchers collect data efficiently and economically.
- Proper sampling improves the accuracy and reliability of research findings.
- Sampling techniques are broadly classified into two categories.
  - **Probability Sampling** – every population element has a known chance of selection.
  - **Non-Probability Sampling** – selection probability is unknown.
- The choice of technique depends on research objectives and available resources.

Preparing Research Design: Sampling Techniques

Probability Sampling

- Probability sampling gives every population element a known and equal chance of selection.
- It reduces selection bias and increases representativeness.
- Results can be generalized to the entire population.
- Statistical analysis is more reliable with this method.
- Commonly used in scientific and quantitative research.
- Requires a complete list of the population.
- Includes **Simple Random**, **Systematic**, **Stratified**, and **Cluster Sampling**.

Preparing Research Design: Sampling Techniques

Simple Random Sampling

- Each population element has an equal chance of being selected.
- Selection is completely random and unbiased.
- Common methods include lottery, random number tables, and computer-generated randomization.
- It is easy to understand and implement for small populations.
- Statistical analysis is straightforward.
- Difficult and costly for very large populations.

Example: Randomly selecting 100 students from a university database.

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Preparing Research Design: Sampling Techniques

Systematic Sampling

- Participants are selected at regular intervals from a population list.
- A sampling interval is calculated using Population Size ÷ Sample Size.
- Example: For 1,000 individuals and a sample of 100, every 10th person is selected.
- The first element is usually selected randomly.
- Easy to implement and manage.
- Saves time compared to random sampling.
- May introduce bias if a hidden pattern exists in the population list.

Preparing Research Design: Sampling Techniques

Stratified Sampling

- The population is divided into homogeneous groups called strata.
- Each stratum shares similar characteristics.
- Samples are selected separately from each stratum.
- Ensures representation of all important subgroups.
- Improves the accuracy and precision of results.
- Requires detailed information about the population.
- Example: Selecting students from Science, Management, and Humanities faculties.

Preparing Research Design: Sampling Techniques

Cluster Sampling

- The population is divided into clusters or groups.
- Entire clusters are selected randomly for the study.
- Individuals within selected clusters become part of the sample.
- Useful for geographically dispersed populations.
- Reduces travel and data collection costs.
- Easier to conduct than individual random sampling.

Example: Selecting several colleges instead of individual students.

Basis	Stratified Sampling	Cluster Sampling
Definition	Population is divided into **homogeneous groups (strata)** based on similar characteristics.	Population is divided into **heterogeneous groups (clusters)**, usually based on location or natural groupings.
Purpose	To ensure representation from all important subgroups.	To reduce cost and simplify data collection.
Selection Method	Samples are selected from **every stratum**.	Only **some clusters** are selected, and data are collected from those clusters.
Group Characteristics	Members within each stratum are similar.	Each cluster contains a mix of different types of individuals.
Representation	All strata are represented in the sample.	Not all clusters need to be represented.
Accuracy	Generally provides higher accuracy and precision.	Usually less precise than stratified sampling.
Cost and Time	More expensive and time-consuming.	More economical and easier to implement.
Example	Dividing students into Science, Management, and Humanities, then sampling from each group.	Selecting a few colleges and surveying all students within those colleges.

Preparing Research Design: Sampling Techniques

Non-Probability Sampling

- The probability of selection for each population element is unknown.
- Selection depends on accessibility, judgment, or other non-random criteria.
- Faster and less expensive than probability sampling.
- Commonly used in exploratory and qualitative research.
- Results may not represent the entire population accurately.
- Greater risk of sampling bias.
- Includes **Convenience**, **Judgment**, **Quota**, and **Snowball** Sampling.

Preparing Research Design: Sampling Techniques

Convenience Sampling

- Participants are selected based on availability and accessibility.
- It is the simplest sampling technique.
- Requires minimal time and resources.
- Suitable for preliminary or pilot studies.
- Easy to conduct in schools, colleges, and workplaces.
- High risk of bias due to non-random selection.

Example: Surveying nearby students on campus.

Preparing Research Design: <u>Sampling Techniques</u>

Judgment Sampling

- Participants are selected based on the researcher's expertise and judgment.
- Researchers choose individuals who can provide valuable information.
- Useful when specialized knowledge is required.
- Often applied in expert interviews and case studies.
- Saves time by focusing on relevant respondents.
- Subjective selection may introduce bias.

Example: Selecting cybersecurity experts for interviews.

Preparing Research Design: <u>Sampling Techniques</u>

Quota Sampling

- The population is divided into categories or groups.
- A specific quota is assigned to each category.
- Participants are selected until each quota is filled.
- Ensures representation of important groups.
- Faster and cheaper than probability sampling.
- Selection within groups is not random.

Example: Selecting equal numbers of male and female respondents.

Preparing Research Design: Sampling Techniques

Snowball Sampling

- Initial participants help identify additional participants.
- The sample grows through referrals and recommendations.
- Useful for hard-to-reach or specialized populations.
- Helps researchers access hidden communities.
- Cost-effective and practical for rare populations.
- May result in biased samples due to social connections.

Example: Research involving hackers or specialized professionals.

Preparing Research Design: Instrument Selection

Instrument Selection

- Instrument selection is the process of choosing appropriate tools for data collection.
- Research instruments help gather information from respondents or observations.
- The choice of instrument affects data quality and research outcomes.
- A suitable instrument ensures valid and reliable data collection.
- Different research objectives require different instruments.
- Instrument selection should align with the research design.

Common instruments include **questionnaires**, **interviews**, **observations**, **schedules**.

Preparing Research Design: Instrument Selection

Questionnaire

- A questionnaire is a set of written questions answered by respondents.
- It is one of the most widely used data collection tools.
- Can be distributed online, by mail, or in person.
- Suitable for collecting information from large populations.
- Responses can be easily organized and analyzed.
- Often used in survey-based research.

Example: Google Forms survey on social media addiction.

Preparing Research Design: <u>Instrument Selection</u>

Types of Questionnaire

- **Open-ended Questionnaire** – Allows respondents to answer in their own words.
- **Closed-ended Questionnaire** – Provides predefined response options. (Type of questions)
- **Structured Questionnaire** – Contains fixed and standardized questions. (Overall questionnaire process)
- **Semi-Structured Questionnaire** – Combines fixed and flexible questions.

Preparing Research Design: Instrument Selection

Questionnaire

Advantages

- Suitable for large populations.
- Cost-effective data collection method.
- Easy to distribute online.
- Facilitates statistical analysis.

Disadvantages

- Low response rates may occur.
- Questions may be misunderstood by respondents.
- Limited opportunity for clarification.

Preparing Research Design: Instrument Selection

Interview

- An interview involves direct communication between researcher and respondent.
- It allows collection of detailed and in-depth information.
- Researchers can clarify unclear responses immediately.
- Interviews can be conducted face-to-face, online, or by phone.
- Suitable for qualitative research.
- Encourages rich and descriptive responses.

Example: Interviewing software developers about agile practices.

Preparing Research Design: Instrument Selection

Types of Interview

- **Structured Interview** – Uses predetermined questions.
- **Semi-Structured Interview** – Combines fixed and open-ended questions.
- **Unstructured Interview** – Flexible conversation with minimal predefined questions.

Preparing Research Design: Instrument Selection

Interview

Advantages

- Provides detailed and comprehensive information.
- Allows clarification of questions and responses.
- Higher response quality than questionnaires.

Disadvantages

- Time-consuming to conduct.
- Requires trained interviewers.
- Possibility of interviewer bias.
- More expensive than surveys.

Preparing Research Design: Instrument Selection

Observation

- Observation involves systematic watching and recording of behaviors or events.
- Data are collected in real-time settings.
- Useful when participants cannot accurately report behavior.
- Common in social and behavioral research.
- Provides direct evidence of activities.
- Can be conducted in natural or controlled environments.
- Helps understand actual behavior rather than reported behavior.

Preparing Research Design: Instrument Selection

Types of Observation

- **Participant Observation** – Researcher actively participates in the setting.
- **Non-Participant Observation** – Researcher observes without involvement.
- **Structured Observation** – Uses predefined observation criteria.
- **Unstructured Observation** – Flexible and exploratory observation.

Preparing Research Design: <u>Instrument Selection</u>

Observation – Advantages & Disadvantages

Advantages

- Collects real-time and natural data.
- Useful for studying behavior and interactions.

Disadvantages

- Observer bias may affect findings.
- Ethical concerns regarding privacy.
- Data recording can be challenging.
- Time-intensive in some studies.

Preparing Research Design: Instrument Selection

Schedule

- A schedule is similar to a questionnaire.
- Responses are recorded by the researcher instead of respondents.
- Useful when respondents have literacy limitations.
- Ensures complete and accurate responses.
- Allows clarification during data collection.
- Produces higher response rates.
- Commonly used in field surveys.

Preparing Research Design: Instrument Selection

Schedule

Advantages

- Higher response accuracy.
- Suitable for illiterate respondents.
- Reduces missing responses.
- Better control over data collection.

Disadvantages

- More expensive than questionnaires.
- Requires trained field workers.
- Time-consuming to administer.

Preparing Research Design: <u>Instrument Selection</u>

Factors Affecting Instrument Selection

- Nature of the research problem.
- Type of data required.
- Characteristics of respondents.
- Available time and budget.
- Required accuracy and reliability.
- Research design and methodology.
- Ethical and practical considerations.

Preparing Research Design: <u>Qualitative Research Methods</u>

- Qualitative research methods are widely used in Computer Applications (CA), Human-Computer Interaction (HCI), Information Systems, Software Engineering, and User Experience (UX) research to understand human behavior, perceptions, experiences, and interactions with technology.
- Unlike quantitative methods that focus on numerical measurements, qualitative methods emphasize understanding the meaning behind user actions and experiences.

Preparing Research Design: Qualitative Research Methods

Qualitative research helps researchers answer questions such as:

- How do users interact with a software system?
- Why do users prefer one application over another?
- What challenges do users face while using a technology?
- How can software products be improved based on user experiences?

Common qualitative methods used in Computer Applications include:

- Diaries
- Case Studies
- Interviews
- Focus Groups

These methods provide rich and detailed insights into user behavior and technology usage.

Preparing Research Design: <u>Qualitative Research Methods</u>

Diaries

The diary method is a qualitative data collection technique in which participants record their experiences, activities, observations, and feelings over a period of time.

Participants maintain a diary or logbook and regularly document:

- their interactions with technology,
- problems encountered,
- user experiences,
- emotional responses,
- usage patterns.

Preparing Research Design: Qualitative Research Methods

Characteristics of Diary Method

- Data are collected continuously.
- Records are maintained by participants themselves.
- Captures real-life experiences.
- Useful for long-term studies.
- Provides detailed contextual information.

Preparing Research Design: Qualitative Research Methods

Advantages

- Captures real-time experiences.
- Reduces memory bias.
- Provides rich qualitative data.
- Useful for longitudinal studies.

Limitations

- Participants may forget to record information.
- Time-consuming.
- Data may be incomplete.
- Requires participant commitment.

Preparing Research Design: Qualitative Research Methods

Case Studies

- A case study is an in-depth investigation of a particular individual, organization, system, project, event, or phenomenon within its real-life context.
- The objective is to gain a comprehensive understanding of a specific case rather than studying large populations.
- Case studies are particularly useful when researchers want detailed insights into complex technological systems.

Preparing Research Design: Qualitative Research Methods

Characteristics

- Focuses on a single case or a small number of cases.
- Provides detailed analysis.
- Uses multiple sources of data.
- Examines real-world situations.

Preparing Research Design: <u>Qualitative Research Methods</u>

Advantages

- Provides deep understanding.
- Suitable for complex problems.
- Uses multiple data sources.
- Produces practical insights.
-

Limitations

- Difficult to generalize findings.
- Time-intensive.
- Researcher bias may occur.

Preparing Research Design: Qualitative Research Methods

Interviews

- An interview is a qualitative data collection method involving direct communication between the researcher and participants.

Researchers ask questions to obtain:

- opinions,
- experiences,
- perceptions,
- motivations,
- and suggestions.

Interviews allow participants to explain their views in detail.

Preparing Research Design: <u>Qualitative Research Methods</u>

Advantages

- Provides detailed information.
- Allows clarification of responses.
- Flexible and interactive.
- Captures emotions and opinions.

Limitations

- Time-consuming.
- Requires interviewer skills.
- Possibility of interviewer bias.
- Data analysis can be complex.

Preparing Research Design: <u>Qualitative Research Methods</u>

Focus Groups

- A focus group is a moderated group discussion involving a small number of participants who share their opinions and experiences regarding a particular topic.
- The researcher acts as a facilitator and encourages interaction among participants.
- Focus groups are useful for exploring collective perceptions and attitudes.

Preparing Research Design: <u>Qualitative Research Methods</u>

Characteristics

- Usually consists of 6–12 participants.
- Guided by a moderator.
- Interactive discussion.
- Generates diverse viewpoints.
- Useful for exploratory research.

Preparing Research Design: Qualitative Research Methods

Advantages

- Collects diverse opinions quickly.
- Encourages interaction.
- Generates new ideas.
- Cost-effective.

Limitations

- Dominant participants may influence discussion.
- Sensitive topics may be difficult to discuss.
- Analysis can be challenging.

Preparing Research Design: <u>Qualitative Research Methods</u>

Ethnography

- Ethnography is a qualitative research approach originally developed in anthropology. It involves studying people, groups, or communities in their natural environment to understand their behaviors, cultures, interactions, and practices.
- In Computer Applications and HCI research, ethnography helps researchers understand how users interact with technology in real-world settings.

Preparing Research Design: Qualitative Research Methods

Ethnographic research involves a continuous cycle of:

- Interviewing
- Observing
- Analyzing
- Repeating
- Theorizing

Preparing Research Design: Qualitative Research Methods

Interviewing

- Researchers communicate with participants to understand:
- experiences,
- motivations,
- beliefs,
- and technology usage patterns.

Interviews provide contextual explanations for observed behaviors.

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Preparing Research Design: <u>Qualitative Research Methods</u>

Observing

Researchers observe participants in their natural environment.

Observation focuses on:

- user actions,
- workflows,
- interactions,
- and challenges.

Example: Observing employees using an ERP system during daily operations.

Preparing Research Design: <u>Qualitative Research Methods</u>

Analyzing

- Collected data are systematically examined.

<u>Researchers</u>:

- code information,
- identify themes,
- detect patterns,
- and interpret meanings.

Preparing Research Design: <u>Qualitative Research Methods</u>

Repeating

- Ethnographic research is iterative.

Researchers repeatedly:

- observe,
- interview,
- analyze,
- and refine understanding.

This process improves accuracy and depth.

Preparing Research Design: <u>Qualitative Research Methods</u>

Theorizing

- Researchers develop explanations and theories based on findings.

Theories help explain:

- user behavior,
- technology adoption,
- organizational practices,
- and social interactions.

Preparing Research Design: <u>Qualitative Research Methods</u>

Advantages of Ethnography

- Provides deep contextual understanding.
- Captures natural behavior.
- Reveals hidden practices.
- Produces rich qualitative data.

Preparing Research Design: Qualitative Research Methods

Limitations of Ethnography

- Time-intensive.
- Requires extensive fieldwork.
- Difficult to generalize findings.
- Researcher bias may occur.

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Preparing Research Design: Qualitative Research Methods

Example of Ethnography in Computer Applications

- **Research Topic:** "How Software Developers Collaborate in Agile Teams"

  Researchers spend several months observing development teams and:
  - conducting interviews,
  - attending meetings,
  - analyzing communication practices.

  The goal is to understand collaboration behavior in software development environments.

Preparing Research Design: Automated Data Collection Methods

- Modern Computer Applications research increasingly uses automated methods for collecting large amounts of data efficiently and accurately.
- Automated data collection refers to the use of software tools, sensors, and computer systems to gather data without continuous human intervention.

These methods are especially important in:

- Artificial Intelligence,
- Data Mining,
- Human-Computer Interaction,
- Cybersecurity,
- Internet of Things (IoT),
- and Big Data Analytics.

Types of Automated Data Collection Methods

System Logs

- Computer systems automatically record user activities.
- **Examples**
  - Login history
  - Application usage
  - Error logs
  - Server activity

Types of Automated Data Collection Methods

Clickstream Data Collection

- Tracks user navigation behavior on websites.

Information Collected

- Pages visited
- Time spent
- Click patterns
- Navigation paths

Types of Automated Data Collection Methods

Sensor-Based Data Collection

- Uses sensors to gather information automatically.
- **Examples**
- GPS sensors
- Motion sensors
- Wearable devices
- IoT sensors

Types of Automated Data Collection Methods

Screen Recording and User Tracking

Software records:

- mouse movements,
- keyboard inputs,
- screen interactions.
- Used in usability studies.

Types of Automated Data Collection Methods

Web Scraping

- Automated extraction of data from websites.

Applications

- social media analysis,
- market research,
- sentiment analysis.

Types of Automated Data Collection Methods

Online Analytics Tools

Tools such as:

- Google Analytics,
- Firebase Analytics,
- Mixpanel
- collect user behavior data automatically.

Types of Automated Data Collection Methods

Advantages of Automated Data Collection

1. High Accuracy

Reduces human errors.

2. Large Data Volume

Handles massive datasets efficiently.

3. Real-Time Collection

Data are collected continuously.

4. Cost Effective

Reduces manual effort.

5. Objective Data

Minimizes researcher bias.

Types of Automated Data Collection Methods

Limitations

1. Privacy Concerns

User consent may be required.

2. Technical Complexity

Requires specialized tools.

3. Data Security Issues

Sensitive information must be protected.

4. Limited Contextual Information

Behavior is recorded, but motivations may remain unknown.